

6. [Maximum mark: 7]

Consider the function $f(x) = 4 \cot x + \sin x$, where $0 < x < \pi$.

(a) (i) Write $f(x)$ in terms of $\sin x$ and $\cos x$.

(ii) Hence or otherwise, sketch the graph of $y = f(x)$, showing the value of the x -intercept. [3]

(b) Find the value of $f^{-1}(2)$. [1]

It is given that $\sec \alpha = 1.5$, where $0 < \alpha < \pi$.

(c) Find the value of $f(\alpha)$. [3]

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

